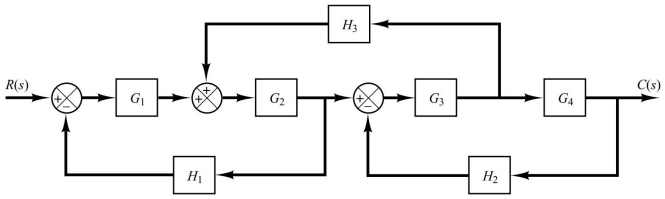
MAS508-G Lab 4 CW37 Assignment

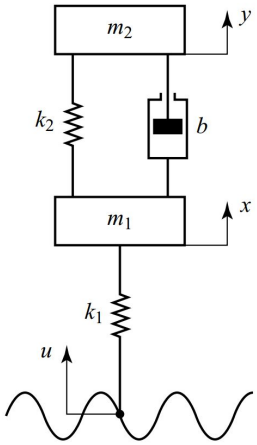
PROBLEM 1

Task: Obtain the closed-loop transfer function $C(s)/R(s)$ by simplying the block diagram of the SISO system shown below.



PROBLEM 2

Task: Consider the following mechanical system with input u as a displacement input.

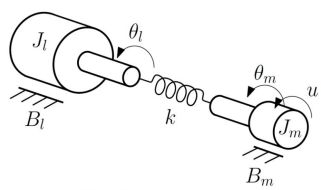


a) Write the equations of motion for the masses m_1 and m_2.
b) Obtain the transfer function for $Y(s)/U(s)$.
c) Simulate the system and show the step response (for t=10 s). Assume the model parameters $m_1 = 100\ kg$, $m_2 = 25\ kg$, $b = 1000\ Ns/m$, $k_1 = 25000\ N/m$ and $k_2 = 3000\ N/m$.

PROBLEM 3

Task: Consider the robot joint with elasticity, a 4th-order system. From the given model equations derive the state-space model of the system.

Hint: State-vector $x = \begin{bmatrix}\theta_l & \dot{\theta}_l & \theta_m & \dot{\theta}_m\end{bmatrix}^T$ and the output $y = \theta_l$.



$u\ [Nm]$ - input torque
$J_m\ [kgm^2]$ - moment of inertia of the motor
$\theta_m\ [rad]$ - angular position of the motor
$B_m\ [Nm/(rad/s)]$ - viscous damping of the motor
$k\ [Nm/rad)]$ - torsional elasticity constant
$J_l\ [kgm^2]$ - moment of inertia of the robot joint
$\theta_l\ [rad]$ - angular position of the robot joint
$B_l\ [Nm/(rad/s)]$ - viscous damping of the robot joint

Model equations:

$$J_l\ddot{\theta}_l + B_l\dot{\theta}_l + k(\theta_l - \theta_m) = 0$$

$$J_m\ddot{\theta}_m + B_m\dot{\theta}_m - k(\theta_l - \theta_m) = u$$

model parameters (SI units)

$J_m = 2$
$B_m = 0.5$
$J_l = 10$
$B_l = 1$
$k = 100$

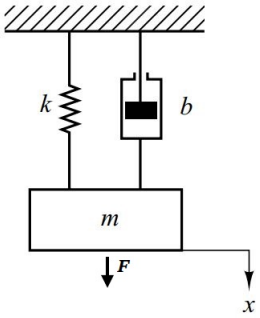
PROBLEM 4

Task: Consider a first-order system with transfer function $G(s) = \frac{80}{s+40}$.

a) Represent the transfer function in the general form and find the time constant T_c, the gain K.
b) Write the step unit input response $c(t)$ (zero initial conditions) in general terms, indicating the forced and natural parts of the solution.
c) Plot the step response of the system.
d) Calculate the rise time T_r and the settling time T_s (using the $\pm 2\%$ criteria).

PROBLEM 5

Task: Consider the following mass-spring-damper system with parameters $m = 2\ kg$, $b = 8\ N.s/m$ and $k = 100\ N/m$ and an external force F.



a) Write down the transfer function $X(s)/F(s)$.
b) Calculate the system's damping ratio and determine the kind of response.
c) Write the general form of the step response $c(t)$.
d) Calculate the peak time T_p, the rise time T_r, the percent overshoot $\%OS$ and the settling time T_s.